

Amelia Eckard

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Education

University of North Carolina at Charlotte <i>M.S. in Artificial Intelligence / Early Entry Program</i>	Charlotte, NC <i>Expected May 2028</i>
University of North Carolina at Charlotte <i>B.S. in Computer Science / Concentration: AI, Robotics, & Gaming</i>	Charlotte, NC <i>Expected May 2027</i>

Experience

Undergraduate Researcher <i>UNC Charlotte, Dr. Todd Dobbs</i>	Charlotte, NC <i>August 2025 – May 2026</i>
- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues. - Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.	
Research Intern <i>Stevens Institute of Technology, Dr. Jina Huh-Yoo</i>	Remote <i>June 2026 – Present</i>
Engineered <i>rq-system</i>/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers.	

Projects

Hera's Garden <i>Node.js, Express, MariaDB, Discord.js, Java, Paper API, Railway</i>	2024 – Present
- Built and maintain a persistent creative-community platform spanning a full-stack website, Discord automation, and a custom Minecraft server ecosystem with scheduled releases, rank-gated access, OAuth2 administration, moderation, and persistent progression systems. - Designed Java/Paper systems for claims, territories, governments, housing, shops, mail, events, NPC societies, and economy features while integrating identity and progression across web, Discord, and game services.	
Vision Lab: ASCII Reconstruction <i>JavaScript, HTML/CSS, Image Processing</i>	2026 – Present
- Developed an image-to-ASCII reconstruction experiment for a web-based computer-vision lab connected to ameliaeckard.com. - Designed reconstruction logic around luminance mapping, local contrast, detail preservation, and edge-aware character selection to retain recognizable image structure.	
Iris <i>Electron, Node.js, Discord.js, SQLite, sqlite-vec, Ollama, MariaDB</i>	2025 – Present
- Developed a local-first personal-memory and community-automation system combining desktop tooling, persistent memory, local language models, and Discord operations. - Built verification, moderation, private-channel, XP/ranking, and notification workflows using SQLite/sqlite-vec and Ollama-based local retrieval and model components.	
RoomCode <i>Python, Flask, JavaScript</i>	2026 – Present
- Built a multi-session collaborative coding environment with real-time synchronization, host-controlled execution, and isolated file trees. - Deployed the platform across 20+ lab sessions with concurrent users, supporting collaborative programming and classroom workflows.	

Technical Skills

Languages	JavaScript, HTML/CSS, Python, Java, Swift
Creative Tech	ARKit, RealityKit, OpenCV, Core ML, image processing
Web/Systems	Node.js, Express, Flask, SQLite, MariaDB, Git, Railway

Certifications & Involvement

Formal Committee Head, Alpha Omega Epsilon – Beta Kappa	August 2025 – May 2026
Coordinated formal-event planning, logistics, communication, and committee execution for the professional and social sorority.	
Volunteer Judge & Field Reset, FIRST Robotics	2025 – Present